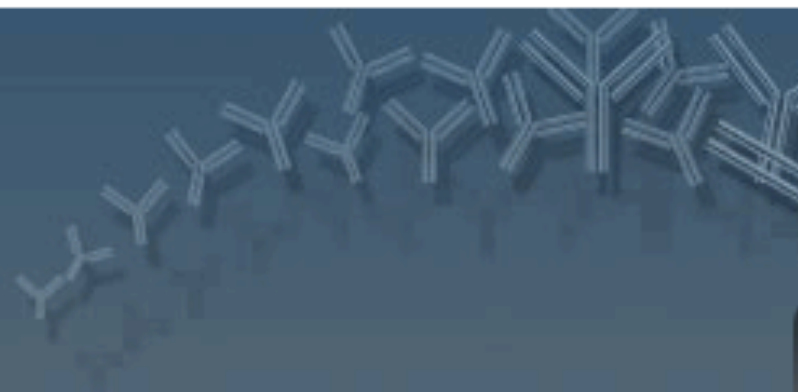






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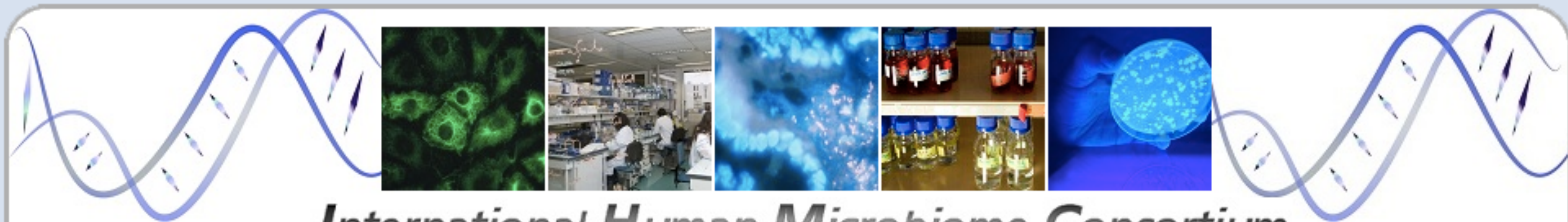
NATURE IMMUNOLOGY | ARTICLE



# Lymphotoxin regulates commensal responses to enable diet-induced obesity

Vaibhav Upadhyay, Valeriy Poroyko, Tae-jin Kim, Suzanne Devkota, Sherry Fu, Donald Liu, Alexei V Tumanov, Ekaterina P Koroleva, Liufu Deng, Cathryn Nagler, Eugene B Chang, Hong Tang & Yang-Xin Fu

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*International Human Microbiome Consortium*

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NATURE | LETTER



# Diet rapidly and reproducibly alters the human gut microbiome

Lawrence A. David, Corinne F. Maurice, Rachel N. Carmody, David B. Gootenberg, Julie E. Button, Benjamin E. Wolfe, Alisha V. Ling, A. Sloan Devlin, Yug Varma, Michael A. Fischbach, Sudha B. Biddinger, Rachel J. Dutton & Peter J. Turnbaugh

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# Metagenomic systems biology of the human gut microbiome reveals topological shifts associated with obesity and inflammatory bowel disease

Sharon Greenblum<sup>a</sup>, Peter J. Turnbaugh<sup>b</sup>, and Elhanan Borenstein<sup>a,c,d,1</sup>

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- Hardening of the arteries
- Inflammatory bowel diseases
- Malnutrition
- Obesity



Lymphotoxin  $\alpha$  enables diet-induced

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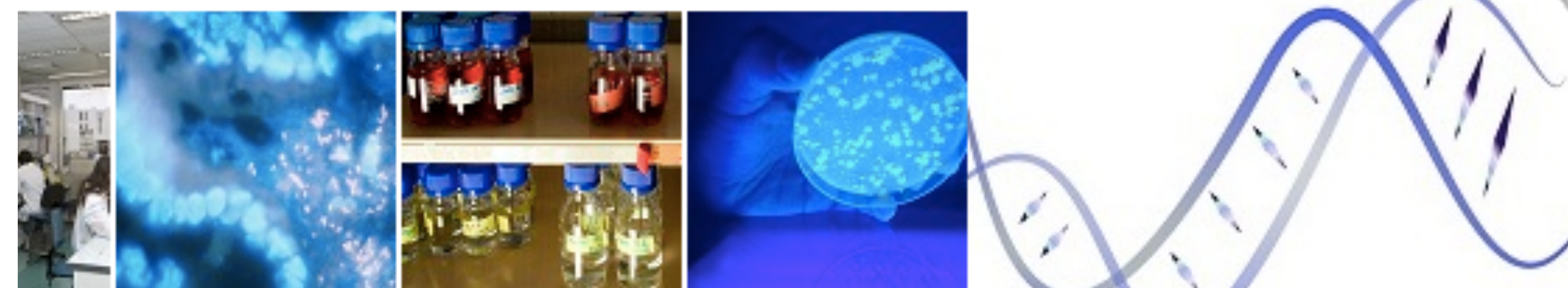
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# Bacterial cells can help tumors

## REPORT

## Potential role of intratumor bacteria in mediating tumor resistance to the chemotherapeutic drug gemcitabine

Leore T. Geller<sup>1,\*</sup>, Michal Barzily-Rokni<sup>2,\*</sup>, Tal Danino<sup>3,†</sup>, Oliver H. Jonas<sup>4,5</sup>, Noam Shental<sup>6</sup>, Deborah Nejman<sup>1</sup>, Nancy Gavert<sup>1</sup>, Yaara Zwang<sup>1</sup>, Zachary A. Cooper<sup>7,8,‡</sup>, Kevin Shee<sup>2</sup>, Christoph A. Thaiss<sup>9</sup>, Alexandre Reuben<sup>8</sup>, Jonathan Livny<sup>2</sup>, Roi Avraham<sup>10</sup>, Dennie T. Frederick<sup>11</sup>, Matteo Ligorio<sup>12</sup>, Kelly Chatman<sup>13</sup>, Stephen E. Johnston<sup>2</sup>, Carrie M. Mosher<sup>2</sup>, Alexander Brandis<sup>14</sup>, Garold Fuks<sup>15</sup>, Candice Gurbatri<sup>16</sup>, Vancheswaran Gopalakrishnan<sup>8</sup>, Michael Kim<sup>8</sup>, Mark W. Hurd<sup>17</sup>, Matthew Katz<sup>8</sup>, Jason Fleming<sup>8</sup>, Anirban Maitra<sup>18</sup>, David A. Smith<sup>2</sup>, Matt Skalak<sup>3</sup>, Jeffrey Bu<sup>3</sup>, Monia Michaud<sup>19</sup>, Sunia A. Trauger<sup>13</sup>, Iris Barshack<sup>20,21</sup>, Talia Golan<sup>21,22</sup>, Judith Sandbank<sup>21</sup>, Keith T. Flaherty<sup>12</sup>, Anna Mandinova<sup>2,23</sup>, Wendy S. Garrett<sup>2,19,24</sup>, Sarah P. Thayer<sup>25</sup>, Cristina R. Ferrone<sup>26</sup>, Curtis Huttenhower<sup>2,27</sup>, Sangeeta N. Bhatia<sup>2,28,29,30,31,32,33</sup>, Dirk Gevers<sup>2,§</sup>, Jennifer A. Wargo<sup>7,8</sup>, Todd R. Golub<sup>34,35,36,¶</sup>, Ravid Straussman<sup>1,¶,¶</sup>

“Certain bacteria express enzymes capable of metabolizing the cancer chemotherapeutic drug *gemcitabine* into an inactive form...an effect that was reversed by antibiotic treatment in mice. A high percentage of human pancreatic ductal adenocarcinomas contain the culprit bacteria (gammaproteobacteria). ...Efficacy of an existing therapy for this lethal cancer might be improved by co-treatment with antibiotics.”